

I. Personal and study details

Student's name: **Majer Tomáš** Personal ID number: **499340**
Faculty / Institute: **Faculty of Electrical Engineering**
Department / Institute: **Department of Computer Science**
Study program: **Open Informatics**
Specialisation: **Data Science**

II. Master's thesis details

Master's thesis title in English:

Learning Structured Prediction Models for Solving Sudoku Puzzles

Master's thesis title in Czech:

Učení strukturních predikčních modelů pro řešení Sudoku

Name and workplace of master's thesis supervisor:

Ing. Vojtěch Franc, Ph.D. Machine Learning FEE

Name and workplace of second master's thesis supervisor or consultant:

Date of master's thesis assignment: **14.02.2026**

Deadline for master's thesis submission: _____

Assignment valid until: **by the end of summer semester 2026/2027**

Jiří Vokřínek
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Head of department's signature

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Vice-dean's signature on behalf of the Dean

III. Assignment receipt

The student acknowledges that the master's thesis is an individual work.
The student must produce his thesis without the assistance of others, with the exception of provided consultations.
Within the master's thesis, the author must state the names of consultants and include a list of references.

Date of assignment receipt
24.02.2026

Bc. Majer Tomáš

Student's signature

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Guidelines:

Design a method for training a Sudoku puzzle solver from examples of puzzle assignments and their corresponding solutions. The solver should support both symbolic inputs (numerical grid representations) and visual inputs (images of Sudoku puzzles). The core of the solver will be a neural network whose final decision layer is a Markov Network predictor. An efficient learning algorithm for this prediction model must be designed and implemented. The resulting system will be empirically evaluated and compared against suitable baseline approaches.

Requirements

- Become familiar with methods for learning neural networks and Markov networks.
- Design and implement a Python library for training and evaluating the proposed predictors.
- Create a benchmark dataset of Sudoku puzzles with ground-truth solutions, covering both symbolic and visual inputs.
- Implement straightforward baseline solutions, for example: a two-stage system consisting of single-digit OCR followed by a hard-coded Sudoku solver.
- Apply the proposed system to the Sudoku benchmark and perform a statistical evaluation of the results, including comparison with baseline methods.

Bibliography / sources:

1. Franc, V., Yermakov, M. Learning Maximum Margin Markov Networks from Examples with Missing Labels. ACML 2021.
2. Franc, V., Průša, D., Yermakov, M. Consistent and Tractable Algorithm for Markov Network Learning. ECML 2022.
3. Wang, P., Donti, P., Wilder, B., Kolter, J. SATNet: Bridging Deep Learning and Logical Reasoning Using a Differentiable Satisfiability Solver. ICML, 2019.